GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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Abstract. A finite causal observer must implement its predictions and its acquisition policy through the same retained state. We prove a stationary acquired-geometry transfer theorem for prepared positive instruments with a finite executable predictive realization. Feedback-uniform overlap yields an average-cost Bellman bias; an invariant unnormalized-ensemble correspondence then constructs one globally calibrated, time-homogeneous finite-state controller, including zero-mass labels and tied optimal actions. An independently charged label erasure, distinct from a physical reset, gives convergence from every admissible preparation under a separate averaged-instrument minorization. Actual acquired density and Bellman curvature yield matching excess risk of order $M^{-2/d}$ against the full-history average predictive-control optimum, with quantitative finite-acquisition entrance bounds and an all-Borel-controller converse. Active Gaussian hidden-state observations and noncommuting quantum instruments verify the same theorem. Visible affine strata give a weighted singular profile. We prove stationary perturbation and finite-precision certificates, preserving the state budget and separating program, workspace, clock and acquisition resources. Protected random audit times are specified operationally; no internally clocked fixed-deadline result is extended by implication.

1. The stationary realization problem

The predictive geometry of an experiment is determined by tests that can actually be performed after its observed histories. Its finite-resource theory requires a second step: predictions must be propagated by an executable causal observer. A static code for a posterior distribution does not supply such an observer. In a controlled experiment the same retained label also selects the next acquisition, changing the distribution that is to be compressed. An indefinitely reused code must therefore solve a global calibration problem, not a separate quantization problem at each date.

We begin with positive experiment kernels, realizable mixtures and a finite separating family of executable future tests. The normalized predictive update is obtained from affine unnormalized report and test responses. This finite realization is an explicit hypothesis, verified from primitive kernels in the applications; finite physical dimension alone is not its proof. The resulting acquired laws depend on actual preparation, reports and actions. No Fisher tensor, covariance geometry or manifold is supplied as a foundational datum.

Our principal result constructs one stationary finite-state observer whose long-run predictive risk has the same memory order as unrestricted checkpoint compression. The full-history comparator optimizes the actions of the same prepared experiment. The proof has four parts. A common acquired mass uniformly over feedback makes the Bellman operator contract in span. The average-cost bias is strongly concave, so its Jensen defect both controls predictive loss from below and admits a second-order acquired-mass upper. A finite-dimensional correspondence on unnormalized label/preparation ensembles closes all calibration equations simultaneously. Finally, a small observer-label erasure and a separate raw physical minorization make that calibrated invariant ensemble attractive from every initial preparation.

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The erasure is important in two distinct ways. It costs one retained label and an explicit risk term, and its rate controls the quantitative entrance time. It neither resets the physical experiment nor supplies a new observation. The fixed-point proof alone would establish a stationary ensemble, not convergence to it. Likewise, a fixed point with real coefficients is not a finite-bit program; the latter is established only under the additional effective certificates in Section 10.

For n possible acquisition cuts and M labels, let $J_*(n)$ be the full-history optimal risk of the protected audit protocol below. The stationary controller gives

$$cM^{-2/d} - \frac{C}{n} \;\le\; \mathcal{R}_{\mathrm{hom}}(n, M) - J_*(n) \;\le\; C\left(M^{-2/d} + \frac{M^{2/d}}{n}\right).$$

Consequently the bounds match for n at least a fixed multiple of $M^{4/d}$, and the long-run excess is comparable to $M^{-2/d}$. The term $M^{2/d}/n$ is an entrance estimate for the constructed observer, not a proved optimal transient cost. An internally enforced exact deadline is a different architecture: it must pay for its phase information. Here an external auditor supplies the stopping request and pays for its own clock. This distinction is part of the theorem, not a convention suppressed in the notation.

Average-cost finite-state approximation, dynamic programming, and value-directed belief compression have substantial prior theories; in particular, near-optimal finite-state controllers are not new as an existence principle [12, 1, 2, 7]. The assertion established here is a quantitative hard-state law through an actual globally calibrated invariant instrument ensemble, together with an all-controller lower, two primitive verifications and separately certified execution. The acquired-overlap, conditional-pooling and curvature mechanisms inherited from [8, 9] are identified in their proofs. The new stationary construction is proved independently of either realization. The theorem remains a sufficient class theorem, not a classification of arbitrary causal experiments.

2. Prepared experiments and the protected task

2.1. **From executable tests to positive instruments.** Fix a time-homogeneous prepared causal experiment on standard Borel spaces, a finite nonempty action set A, and its legal finite future experiments and bounded readouts. A strategy uses only past reports, actions and independent private randomness. Histories are predictively equivalent when they have the same legal action interface and all executable future-test expectations agree.

We require a finite separating affine test realization. More precisely, the constant test and finitely many executable tests give a Borel vector $u(h) \in \mathbb{R}^d$ whose attained image is Borel and has a Borel realization section. Equality of these coordinates is equivalent to equality of all future tests. The unnormalized expectation of every coordinate test jointly with a next report is affine in the incoming coordinates. Realizable random mixtures form a compact convex base $B \subset \mathbb{R}^d$ with nonempty interior in this chart. We write

$$C = \{(w, wu) : w \ge 0,\ u \in B\} \subset \mathbb{R}^{d+1}, \qquad e(w, v) = w.$$

This is a closed pointed generating cone, and e is its normalization functional. The associated base norm is

$$\|z\|_B = \inf\{e(z_+) + e(z_-) : z = z_+ - z_-,\ z_\pm \in C\}.$$

On a finite direct sum of label cones we use the sum base norm.

For each action a let λ_a be a reference probability and $L_a(y)$ a Borel, integrable family of positive linear maps on C, with $\bar{L}_a = \int L_a(y)\lambda_a(dy)$ preserving e. Positivity is positivity on this cone; complete positivity is separately verified for quantum instruments. Put

$$L_a(y)(1, u) = (\ell_a(y, u), U_a(y, u)), \qquad F_a(y, u) = U_a(y, u)/\ell_a(y, u). \tag{1}$$

Both ℓ_a and U_a are affine in u. All identities hold for every incoming state off one reference-null set for each action. At zero likelihood the numerator is zero by positivity and bounded tests, and no normalized state is used on that event. The actual acquired row and its integration operator are

$$\mu_{a,u} = (F_a(\cdot, u))_{\#}(\ell_a(\cdot, u)\lambda_a), \qquad K_a f(u) = \int f(v)\mu_{a,u}(dv). \tag{2}$$

A concrete experiment may first be given on a larger physical state space. Its asserted finite test image and positive maps must then be derived from that experiment, as in Section 7.

Lemma 2.1 (Predictive realization)**.** *Under the preceding hypotheses, the attained test image realizes the predictive quotient as a Borel causal state space. It is minimal among sufficient Borel statistics whose attained images have Borel realization sections. Report and update kernels descend jointly in the finite action coordinate.*

Proof. Equal test coordinates imply equal future tests by the separating realization hypothesis; the converse holds because the coordinates are executable tests. For another sufficient statistic, compose its realization section with $h \mapsto u(h)$. Sufficiency makes this map independent of representatives, proving Borel factorization. The common null sets and the affine instrument identities give Borel descended kernels. Successive integration proves realization of every legal finite continuation. This proves the asserted minimality; it does not assert that an arbitrary measurable quotient is standard Borel. The larger mixture domain B need not be the support of any one acquired law. □

2.2. **Prediction, acquisition and audit.** A fixed bounded executable audit Z has conditional mean u and affine second moment $c(u)$. A response $r \in B$, committed before the audit, has squared loss

$$\mathbb{E}[\|Z - r\|^2 \mid u] = c(u) - 2\langle u, r\rangle + \|r\|^2, \qquad g(u) = c(u) - \|u\|^2. \tag{3}$$

An injective affine probability chart of a categorical or informationally complete measurement gives the same formula on its affine image, with the induced Euclidean norm. Orthogonal components of the actual audit noise remain in c; they are not discarded. Assume the loss is bounded by L_0 on the stated response set.

An infinite unaudited reference execution is used to define costs, not to perform incompatible audits at all times. For an integer $n \geq 1$, an external auditor independently chooses T uniformly from $\{1, \ldots, n\}$, hides T, and requests the response immediately after report T. It then performs the single protected audit and ends the experiment. Before that request the controller sees only the prescribed acquisition reports. The expected risk is the average of the potential single-audit risks at cuts $1, \ldots, n$. For a quantum experiment no intermediate audit is physically inserted.

This protocol uses at most n raw acquisition calls, one initial preparation and one audit, hence a pathwise physical-call budget $N = n + 2$. Its expected raw-report count is $(n + 1)/2$, not n. The auditor pays for its uniform draw and an $\lceil \log_2(n + 1) \rceil$-bit counter. The observer is not given that counter or an impending-last-report signal. This external clock is recorded separately from observer memory and physical time.

Write X_t for the full-history predictive vector on the reference execution. The full-history comparator may use every legal adaptive action strategy, including its own time and workspace; its risk is

$$J_*(n) = \inf_{\pi} \frac{1}{n} \sum_{t=1}^{n} \mathbb{E}_\pi g(X_t). \tag{4}$$

A checkpoint has the same unrestricted acquisition history and, when requested, encodes it into M labels with one label-only decoder. Its acquisition workspace is not identified with M.

A homogeneous finite observer has a single set S of at most M labels, an initial label, a fixed action distribution p_s at each label, a Borel update distribution $\gamma(j \mid s, a, y)$ and a fixed readout r_s.

Its only retained dynamic variable is its label. A fresh action/update coin is independent of the physical state given that label; a retained seed would count as state. Neither a history tape, a real posterior register, a phase counter nor the previous report is available at the next call. The external request reads the existing label and ends the run. Put $\mathcal{R}_{\mathrm{hom}}(n, M) = \inf R_n$ over these programs. Their tables may depend on the known experiment and on M; the controller constructed below works for every n and every initial preparation. Known real table entries are counted as program data, not as costless bit strings.

3. Stationary acquired-geometry transfer

Definition 3.1 (Stationary acquired-overlap class)**.** In the experiment of Section 2, choose compact convex sets $K \subset \operatorname{int} D \subset B$, with nonempty interior, and $r > 0$ such that $K + B_r \subset D$. Assume the following fixed certificates.

Acquired regularity. Every normalized instrument output from D lies in K off the common null sets. For $u \in D$ the likelihood is positive almost everywhere, the acquired row is weakly continuous in u, and

$$\mu_{a,u} \le \Lambda \operatorname{vol}_d|_K, \qquad \mathrm{TV}(\ell_a(\cdot, u)\lambda_a, \ell_a(\cdot, v)\lambda_a) \le L_{\mathrm{rep}}\|u - v\|.$$

These assertions hold for all actions and all incoming states in D, including retained barycenters not produced by an exact filter. K has global Borel grid partitions of at most m cells, with diameter at most $C_K m^{-1/d}$. Their boundaries have zero mass under every acquired row. Convex compact K and the density bound supply such grids.

Feedback-uniform acquired overlap. There are $b \ge 1$, $\epsilon_* > 0$ and a probability ζ on K such that

$$\mathrm{Law}_u^{\pi}(X_b) \ge \epsilon_* \zeta \quad (u \in D,\ \pi \text{ any legal } b\text{-call feedback policy}). \tag{5}$$

Every report and action in the block is included in this actual law.

Raw averaged minorization. For a normalized $z_* \in C$, with predictive coordinate in K, and $\eta_* > 0$,

$$\bar{L}_a z \ \ge_C \ \eta_* e(z) z_* \quad (z \in C,\ a \in A). \tag{6}$$

This is a condition on unnormalized physical instruments, not a contraction of posterior trajectories. It is separate from (5). The instruments are integrable in a fixed finite-dimensional operator norm. All constants are independent of the program, n and M.

Let $p = 2/d$. Denote by $\bar{R}(\gamma, u_0)$ the limsup of the protected average risks R_n of a homogeneous program started at $u_0 \in D$. The next theorem does not presuppose stationary calibration or an optimal finite observer.

Theorem 3.2 (Globally calibrated stationary transfer)**.** *Suppose Definition 3.1 and the executable realization and task hypotheses hold. There are a constant ρ and a continuous concave function h on D satisfying*

$$\rho + h(u) = g(u) + \min_{a \in A} K_a h(u). \tag{7}$$

The function $h + \|\cdot\|^2$ is concave, and, writing $H = \operatorname{osc}_D h$,

$$H \le b \operatorname{osc}_D g/\epsilon_*, \qquad \mathrm{Lip}_K h \le H/r, \qquad |J_*(n) - \rho| \le C/n. \tag{8}$$

For each $M \ge 2$ and each $0 < \alpha \le 1/2$ there is one homogeneous observer $\gamma_{M,\alpha}$ with at most M labels. One label is reserved for independent erasure at probability α after each report; the physical state is not reset. Its actual invariant label/preparation ensemble is calibrated, and its invariant risk obeys

$$\rho + cM^{-p} \ \le \ R_\infty(\gamma_{M,\alpha}) \ \le \rho + C(M-1)^{-p} + \alpha H. \tag{9}$$

The joint physical/label ensemble of this fixed observer contracts in sum base norm by $1-\alpha\eta_*$. *In particular, from every initial preparation in* D *and every initial label,*

$$R_n(\gamma_{M,\alpha}, u_0) \le \rho + C(M-1)^{-p} + \alpha H + \frac{2L_0}{n\alpha\eta_*}. \tag{10}$$

Conversely, every legal Borel finite-state observer with at most M *labels, even one with time-dependent tables, satisfies*

$$R_n \ge \rho + cM^{-p} - C/n. \tag{11}$$

The converse does not grant time-dependent tables to the constructed upper.

Choosing a dyadic α *between* $\frac{1}{4}(M-1)^{-p}$ *and* $\frac{1}{2}(M-1)^{-p}$ *gives*

$$cM^{-p} \le \inf_{\gamma} \bar{R}(\gamma, u_0) - \rho \le CM^{-p}, \tag{12}$$

uniformly in the admissible initial preparation. Moreover

$$cM^{-p} - C/n \le \mathcal{R}_{\mathrm{hom}}(n, M) - J_*(n) \le C(M^{-p} + M^p/n). \tag{13}$$

Thus the finite-acquisition orders match whenever $n \ge C_* M^{2p}$, *for a fixed sufficiently large* C_*. *All constants depend only on the fixed experiment and task certificates, not on* n, M.

The finite-observer upper constructs the action and compression rules through the same actual invariant ensemble. It is not a fixed-policy comparison. Equation (13) gives no matching assertion for its entrance term in the complementary small-n region. It concerns $N = n+2$ as a pathwise call budget for the specified random audit, not a fixed-deadline protocol using exactly n reports. In particular there is no $M-n$ clock subtraction in (12).

Corollary 3.3 (Checkpoint comparison)**.** *For the same protected audit and action class, put* $\nu_n^\pi = n^{-1}\sum_{t=1}^n \mathrm{Law}_\pi(X_t)$ *and* $B_n^\pi = n^{-1}\sum_{t=1}^n \mathbb{E}_\pi g(X_t)$. *Then*

$$\mathcal{R}_{\mathrm{cp}}(n, M) = \inf_\pi\{B_n^\pi + q_M(\nu_n^\pi)\}, \qquad cM^{-p} \le \mathcal{R}_{\mathrm{cp}}(n, M) - J_*(n) \le CM^{-p}, \tag{14}$$

where $q_M(\nu) = \inf_{z_1,\dots,z_M} \int \min_j \|x - z_j\|^2 \nu(dx)$. *The decoder has no free time argument. Its global cell code suffices for the upper.*

The two inequalities (5) and (6) have different jobs. Acquired overlap controls the full-history Bellman problem. Raw averaged minorization makes the single compiled finite observer approach its calibrated ensemble. Neither is silently substituted for the other. The exact conditional identities and the all-observer lower below do not need a stationary starting law.

4. Average-cost bias and acquired curvature

Lemma 4.1 (Block span contraction)**.** *Define* $\mathcal{T}f = g + \min_a K_a f$ *on* $C(D)$. *Under* (5),

$$\mathrm{osc}_D(\mathcal{T}^b f - \mathcal{T}^b q) \le (1-\epsilon_*)\,\mathrm{osc}_D(f-q).$$

There is a solution of (7), *unique modulo constants in* h, *with the bounds in* (8). *It is strongly concave in the stated task chart.*

Proof. For a fixed starting state, compare a b-step policy minimizing the terminal-f objective with one minimizing the terminal-q objective. Finite actions and weak continuity give Borel selectors by finite backward induction; arbitrarily minimizing policies give the same inequalities. Running costs cancel when the same policy is used in this comparison. Thus their difference is bounded above and below by terminal expectations of $f-q$ under legal policies. Every such terminal law is $\epsilon_*\zeta + (1-\epsilon_*)\widetilde{\mu}$. Therefore every value of $\mathcal{T}^b f - \mathcal{T}^b q$ lies between

$$\epsilon_*\zeta(f-q) + (1-\epsilon_*)\inf_D(f-q) \quad \text{and} \quad \epsilon_*\zeta(f-q) + (1-\epsilon_*)\sup_D(f-q).$$

This proves the contraction, including feedback policies with different state-dependent minimizing actions.

The quotient $C(D)/\mathbb{R}$ with oscillation norm is complete. The translation-equivariant $\mathcal{T}$ induces a map whose bth iterate is a strict contraction. Its unique fixed class is also fixed by $\mathcal{T}$, since $\mathcal{T}$ commutes with its iterate. Choose its representative h to vanish at a fixed point of D. Then $\mathcal{T}h = h + \rho$. Since $\operatorname{osc}_D(\mathcal{T}^b 0) \le b \operatorname{osc}_D g$, the contraction gives $H \le b \operatorname{osc}_D g/\epsilon_*$. This is the usual span-contraction argument, applied here to feedback-uniform acquired laws rather than to a latent transition alone [3, 6].

If f is concave, its perspective $wf(v/w)$ is concave on the positive cone of its domain. Hence $K_a f$ is concave by (1). The minimum of finitely many concave functions is concave. Iteration from zero and uniform convergence modulo constants show that h is concave. Equation (7) gives $h - g = \min_a K_a h - \rho$ concave; since $g = c - \|\cdot\|^2$ with affine c, this proves strong concavity without a Hessian assumption.

For a concave function on D, extending the line through two points of K by length r at either end bounds each intermediate slope in absolute value by H/r. Thus h is H/r-Lipschitz on K, including nonsmooth switching surfaces. Bellman minimizers in (7) exist and have a Borel selection. Summing the Bellman inequality along any full-history strategy gives a lower average cost $\rho - H/n$ at cuts $0, \dots, n-1$; a minimizing selector gives an upper $\rho + H/n$. Shifting these cuts to $1, \dots, n$ changes the average by at most $\operatorname{osc}_D g/n$. This proves the asserted bound for $J_*(n)$ and full-history average optimality of ρ. □

Lemma 4.2 (Boundary-valid acquired-curvature estimate)**.** *Let v be concave and L-Lipschitz on compact convex $K \subset \mathbb{R}^d$, and let μ be a probability with density at most Λ. A single grid partition $\mathcal{P}_m$ of K with at most m cells satisfies*

$$\sum_{C \in \mathcal{P}_m} \mu(C) v\left(\frac{\int_C x \, d\mu}{\mu(C)}\right) - \int v \, d\mu \le C_{K,d} \Lambda L m^{-2/d}. \tag{15}$$

Zero-mass cells are omitted. The grid is independent of v, μ; the centroids are not.

Proof. This is the acquired-mass curvature mechanism of [9], recalled to make the stationary proof self-contained. The convex function $-v$ extends to a convex L-Lipschitz function f on $\mathbb{R}^d$: take the supremum of its supporting planes at interior points of K. Interior slopes have norm at most L, and continuity gives equality on the boundary. Its distributional Laplacian $\kappa = \Delta f$ is a positive locally finite measure.

For a cube C of side s and center z, mollify f and subtract its tangent plane at z, obtaining a nonnegative convex w with $w(z) = 0$. On a ball of radius $R = 2\sqrt{d}\, s$, submean estimates bound w on C by a fixed multiple of the average of w on the surrounding sphere. Convex radial slopes give $w(z + R\theta) \le R \partial_R w(z + R\theta)$. Integration and the divergence theorem therefore yield

$$\sup_C w \le C_d R^{2-d} \int_{B_R(z)} \Delta f.$$

For $d = 1$ the sphere consists of the two endpoints and the same calculation applies. Tangent contributions cancel at the conditional centroid, and its value of w is nonnegative. Passing through mollification with a slightly enlarged ball proves

$$\int_C f \, d\mu - \mu(C) f(\mathbb{E}[X \mid C]) \le C_d s^{2-d} \mu(C) \kappa(\overline{B}_{3\sqrt{d}s}(z)).$$

A smooth cutoff on a fixed neighborhood of K bounds its curvature mass by $\int \chi \, d\kappa = -\int \nabla\chi \cdot \nabla f \le L \int |\nabla\chi|$. The enlarged balls have bounded overlap and $\mu(C) \le \Lambda s^d$. Summation gives $C_{K,d} \Lambda L s^2$. Divide a bounding cube into $\lfloor m^{1/d} \rfloor$ parts per coordinate, intersect with K, and discard empty cells. This proves (15), with the case $m = 1$ absorbed by its fixed constant. No positive distance of the acquired support from the boundary of K is used. □

Lemma 4.3 (Actual-law small-ball lower)**.** *Every mixture of the acquired rows in Definition 3.1 satisfies*

$$q_M(\mu) \geq c_d \Lambda^{-2/d} M^{-2/d}, \qquad c_d = \tfrac{1}{2}(2v_d)^{-2/d}, \tag{16}$$

where v_d is the Euclidean unit-ball volume.

Proof. The density bound survives mixing. The union of radius-r_0 balls about M readouts has probability at most $\Lambda M v_d r_0^d$. For $r_0 = (2\Lambda M v_d)^{-1/d}$ at least half the actual mass lies outside. Randomized assignment cannot improve the pointwise minimum over its finite readouts. This is a bound on an acquired law, not on a formally reachable set. □

5. A globally calibrated invariant ensemble

Lemma 5.1 (Stationary calibration with charged erasure)**.** *Fix a grid $\mathcal{P}_m = \{C_1, \dots, C_m\}$ as in Lemma 4.2 and $0 < \alpha \leq 1/2$. There is one Borel homogeneous observer with labels $0, 1, \dots, m$ and an invariant normalized ensemble (z_s) such that its positive-mass labels have their actual conditional predictive means q_s, its action distributions are supported on $\arg\min_a K_a h(q_s)$, and it erases to label 0 independently with probability α after each report. Nonerased updates send $F_a(y, q_s)$ to its common grid cell. Zero-mass labels require neither a conditional mean nor a division by zero.*

Proof. Write $C_K = \{w(1, q) : w \geq 0, q \in K\}$. The set

$$\mathcal{E} = \{(z_0, \dots, z_m) \in C_K^{m+1} : \sum_s e(z_s) = 1\}$$

is compact and convex. At $z_s = w_s(1, q_s)$ with $w_s > 0$, allow every probability vector p_s supported on the finite minimizing set $\arg\min_a K_a h(q_s)$. At $w_s = 0$, allow every p_s and use any $q_s \in K$; the contribution of this row is defined to be zero. For fixed z and an allowed choice p, set

$$\mathcal{F}_0(z, p) = \alpha \sum_{s,a} p_{sa} \bar{L}_a z_s, \tag{17}$$

$$\mathcal{F}_j(z, p) = (1 - \alpha) \sum_{s,a} p_{sa} \int \mathbf{1}_{\{F_a(y,q_s) \in C_j\}} L_a(y) z_s \, d\lambda_a(y), \quad 1 \leq j \leq m. \tag{18}$$

These vectors lie in C_K, and their masses sum to one. Thus $\mathcal{F}$ is a correspondence from $\mathcal{E}$ to itself. For fixed z, the allowed set of action probabilities is a nonempty product of simplices. The displayed map is linear in these probabilities, so its image is convex and compact. This use of all tied minimizers, rather than a discontinuous selected action, is essential.

We verify closedness of the graph. Suppose $z^{(k)} \to z$ and pass to a subsequence on which all action probabilities converge. At positive limiting mass, $q_s^{(k)} \to q_s$. Since $K_a h$ is continuous, a limiting action with positive probability is still minimizing. Outside a common λ_a-null set, the positive linear instrument makes $F_a(y, q_s^{(k)}) \to F_a(y, q_s)$. The acquired law of the latter point charges no grid boundary. Positivity of its likelihood makes the boundary preimage λ_a-null as well. Dominated convergence applies to (18), using the integrable operator norm of L_a and compactness of K. At a zero-mass limiting row, every integrated contribution has norm bounded by a constant times $w_s^{(k)} \to 0$, uniformly in its arbitrary coordinates and actions. Thus no continuity of normalization at zero mass is needed. The graph is closed. A compact-valued closed-graph correspondence on a compact domain is upper semicontinuous.

The finite-dimensional Kakutani theorem [4] gives $z^* \in \mathcal{F}(z^*)$. Fix one witnessing set of action rows p_s. Equations (17)–(18) are exactly the invariant equations of the positive instrument obtained by executing those rows and those report updates. They are not moment constraints detached from an implementable kernel. The prepared label mixture with unnormalized states z_s^* is realizable by the mixture hypothesis. After an update it reproduces the same ensemble. Consequently, at every positive-mass output label its programmed q_j equals the actual conditional predictive mean. The

erasure label has the mean of all outgoing physical states, not a prescribed physical reset state. Arbitrary legal codepoints and action probabilities complete rows that have zero invariant mass, making the program defined from every allowed initial label. □

Proposition 5.2 (Invariant risk and attraction)**.** *The observer of Lemma 5.1 obeys* (9) *and* (10)*. Its invariant ensemble is unique as an ensemble of the finite executable test coordinates. Convergence of this ensemble is not asserted to be total-variation convergence of an unrepresented microscopic state.*

Proof. At stationarity put $Q = q_S$, choose A from p_S, and let $W = F_A(Y, Q)$. Its law is the actual acquired mixture

$$\mu = \sum_{s,a} w_s p_{sa} \mu_{a,q_s}, \qquad \mu \le \Lambda \operatorname{vol}_d.$$

Write Q' for the predictive mean of the next label. Summing (7) over the calibrated incoming ensemble, using minimizing actions and stationarity, gives the exact identity

$$R_\infty - \rho = \mathbb{E}h(Q') - \mathbb{E}h(W) = (1-\alpha)\mathcal{J}_{\mathcal{P}_m}(h,\mu) + \alpha\left[h\left(\int x\, d\mu\right) - \int h\, d\mu\right]. \tag{19}$$

Here $\mathcal{J}$ is the cell-centroid gap in (15). Conditional expectation is justified by the unnormalized instrument equations, including all incoming labels and action rows. The curvature estimate and $\operatorname{osc}_K h \le H$ prove the upper. Since $h + \|\cdot\|^2$ is concave,

$$\mathbb{E}h(Q') - \mathbb{E}h(W) \ge \mathbb{E}\|W - Q'\|^2 \ge q_{m+1}(\mu).$$

The last inequality includes randomized encoders. Lemma 4.3 proves the lower.

For the fixed table, let P_γ be its positive linear map on the direct sum of $m+1$ label cones. The erase row and (6) imply, for every nonnegative ensemble Z,

$$P_\gamma Z \ge \alpha\eta_* e_{\mathrm{tot}}(Z) Z_*, \tag{20}$$

where Z_* places z_* in label 0. Thus $P_\gamma = \gamma_* R + (1-\gamma_*)Q_\gamma$, where $\gamma_* = \alpha\eta_*$, $RZ = e_{\mathrm{tot}}(Z)Z_*$, and Q_γ is positive and normalization preserving. If $\gamma_* = 1$, convergence is immediate; otherwise this formula defines Q_γ. Positive mass-preserving maps contract base norm. On differences with zero total mass the R term vanishes, proving

$$\|P_\gamma^t Z - z^*\|_{\oplus B} \le 2(1-\gamma_*)^t.$$

This proof decomposition does not prescribe that the physical device actually reset at a random time.

With the readouts fixed, expected audit loss is a bounded affine functional of the physical/label ensemble. It lies in $[0, L_0]$ on normalized positive ensembles. The difference of its expectations is at most L_0 times the sum base norm. Summing the geometric estimate over $t = 1, \dots, n$ gives an error at most $2L_0/(n\alpha\eta_*)$. This proves the entrance bound from an arbitrary admissible physical preparation and arbitrary initial label, even if that initial label is not calibrated. □

The fixed point solves a global recursion, not just the equality case of a one-step row quantization. The codepoints at which actions are chosen are exactly the centroids of the invariant executed law. Calibration need not hold during entrance; the separate contraction estimate controls that period. A report that enters the erasure row is still an acquired report and is paid for.

6. The all-controller converse and checkpoint law

Lemma 6.1 (Controlled conditional barycenters)**.** *For any legal finite-state observer, let* $Q_t = \mathbb{E}[X_t \mid S_t]$*. Given* S_t *and the next chosen action* A_t*, put* $W_{t+1} = F_{A_t}(Y_{t+1}, Q_t)$*. Then*

$$\mathbb{E}[X_{t+1} \mid S_t, A_t, Y_{t+1}] = W_{t+1}, \qquad Q_{t+1} = \mathbb{E}[W_{t+1} \mid S_{t+1}].$$

All conditional statements concern their positive-mass events. They hold for Borel randomized updates and for time-dependent tables.

Proof. The fresh action is independent of the physical state conditionally on the retained label. Both the report likelihood and the unnormalized test numerator are affine in that label's conditional predictive state. Consequently their conditional integrals are $\ell_a(y, Q_t)$ and $U_a(y, Q_t)$. Division on positive likelihood gives the first identity; equivalently it follows by integration against bounded measurable report functions, without conditioning a continuous report on a positive-probability singleton. Fresh update randomness carries no further physical information given (S_t, A_t, Y_{t+1}). The tower property proves the second identity. Earlier private randomizations are included in the observer filtration, and no future audit outcome is used. This is the likelihood-reweighted pooling identity of [8, 9]. □

Proposition 6.2 (Bellman loss decomposition for every observer)**.** *At each cut define*

$$\Delta_t = \mathbb{E}\left[K_{A_t} h(Q_t) - \min_a K_a h(Q_t)\right], \qquad J_{t+1} = \mathbb{E}h(Q_{t+1}) - \mathbb{E}h(W_{t+1}).$$

Then $\Delta_t \geq 0$*,* $J_{t+1} \geq cM^{-2/d}$*, and*

$$\mathbb{E}g(Q_t) - \rho = \mathbb{E}h(Q_t) - \mathbb{E}h(Q_{t+1}) + \Delta_t + J_{t+1}. \tag{21}$$

In particular the converse in Theorem 3.2 holds without stationary calibration or an invariant start.

Proof. Substitute the Bellman equation at Q_t and then add and subtract $\mathbb{E}h(Q_{t+1})$. Lemma 6.1 identifies $\mathbb{E}K_{A_t} h(Q_t) = \mathbb{E}h(W_{t+1})$, proving the equality. The conditional state Q_t is a realizable mixture, lies in K for $t \geq 1$, and lies in D at the initial cut. Its actual outgoing law is a mixture of the rows in Definition 3.1, hence has density at most Λ. Strong concavity and conditional pooling give

$$J_{t+1} \geq \mathbb{E}\|W_{t+1} - Q_{t+1}\|^2 \geq q_M(\mathrm{Law}(W_{t+1})) \geq cM^{-2/d}.$$

The label may be randomized and may pool arbitrarily different incoming rows; none of these enlarges its number of reproduction points. This is an information/decision lower against every declared program, not just a chosen common-cell construction.

Sum (21) at $t = 0, \ldots, n-1$. The bias telescope is bounded by H. Shifting the $g(Q_t)$ cuts to $1, \ldots, n$ costs at most $\mathrm{osc}_D\, g/n$. Any actual label readout r_{S_t} satisfies

$$\mathbb{E}\|Z_t - r_{S_t}\|^2 = \mathbb{E}g(Q_t) + \mathbb{E}\|Q_t - r_{S_t}\|^2 \geq \mathbb{E}g(Q_t).$$

This identity means the potential single-audit risk; it does not insert intermediate physical audits. We obtain (11). Proposition 5.2, the dyadic choice of α, and (8) give the remaining assertions of Theorem 3.2. The lower constant in the finite matching region remains positive after choosing C_* sufficiently large. We do not claim that this sufficient entrance scale is necessary. □

Proof of Corollary 3.3. Fix a legal acquisition strategy π. At the hidden uniform cut T, conditional projection gives Bayes risk B_n^π plus the squared distortion of the actual X_T law. The decoder has only its M-valued label, so optimizing its encoder and decoder gives $q_M(\nu_n^\pi)$. Randomization cannot lower the nearest-point cost. Infimizing over π proves the identity. Each acquired row, and hence ν_n^π, has density bounded by Λ on K. Lemma 4.3 gives a uniform lower. A single global cover with M cells and fixed representatives gives a uniform $CM^{-2/d}$ upper, independently of the strategy and of T. Taking arbitrarily minimizing full-history strategies proves both inequalities. In particular, the actual mixture ν_n^π is not replaced by uniform volume or a formal reachable set. □

7. Two raw controlled instruments

The next verifications derive predictive coordinates from executable audits. They do not use a realization to prove Theorem 3.2. Some local acquired-density calculations were established for finite controlled horizons in [9]; we supply them again and prove the additional averaged-cone minorization required for stationary realization.

Theorem 7.1 (Active Gaussian hidden-state observations)**.** *Let the physical state have $q \geq 2$ values. Fix an invertible column-stochastic matrix P with $P_{ij} \geq a_0 > 0$. An acquisition action a chooses one of finitely many Gaussian sensors. Conditional on pre-transition state i, its report has distribution $N(m_i^a, \Sigma_a)$ on $\mathbb{R}^{q-1}$, with Σ_a positive definite and the $(q-1)\times(q-1)$ likelihood-difference matrix of full rank. After the report the physical state transitions by P. The audit reveals the current state, and the task is categorical squared prediction loss. For every fixed such experiment, Theorem 3.2 holds with $d = q-1$. In particular it compares one homogeneous finite controller with all full-history adaptive sensor-selection strategies, not merely with a prescribed passive filter.*

Proof. Let u be the conditional probability vector of the physical state. A categorical audit is executable and separates u. Every legal future experiment has expectation linear in u, so equality of u is exactly future-test equality. Mixture preparations give the full simplex; the coordinate image is Borel, and preparation of each probability vector gives a Borel section. On its affine hull the instrument is

$$L_a(y)u = P\,\mathrm{diag}(f_1^a(y), \dots, f_q^a(y))u,$$

where f_i^a are Gaussian densities. This formula uses Lebesgue measure; dividing all densities by their positive finite mixture gives the reference probability required above, without changing the instrument. The likelihood is $\sum_i u_i f_i^a(y)$ and the normalized update is P times the usual Bayes vector. Thus affine test closure and integrable positive maps are obtained directly from the physical experiment. The audit Brier risk is $g(u) = 1 - \|u\|^2$ in the simplex hyperplane, including its actual sampling noise.

Set $K = P\Delta_{q-1}$ and $D = \{u \in \Delta_{q-1} : u_i \geq a_0/2\}$. Invertibility makes K full-dimensional in the hyperplane. It is compact with a positive buffer inside D, and every report sends D into K. Likelihoods are positive. Report total variation is at most a fixed multiple of $\|u - v\|$ by linearity; dominated integration gives weak continuity.

Before P, use log odds $z_i = \log(w_i/w_q)$ of the Bayes posterior w. They equal

$$z_i = \log(u_i/u_q) + (m_i^a - m_q^a)^T\Sigma_a^{-1}y - \tfrac{1}{2}\big[(m_i^a)^T\Sigma_a^{-1}m_i^a - (m_q^a)^T\Sigma_a^{-1}m_q^a\big].$$

Under the actual mixture report law this is a finite mixture of nonsingular Gaussian distributions. Their means range over a compact set for $u \in D$, and their covariance matrices are fixed for each of finitely many sensors. Hence the log-odds density is at most $C\exp(-c\|z\|^2)$, uniformly in u, a. The inverse Jacobian of the logistic chart is $(\prod_i w_i)^{-1}$, up to the fixed affine-volume normalization, and is at most $C\exp(C\|z\|)$. Gaussian decay therefore bounds the posterior density throughout the simplex, including its approach to every face. The invertible affine map P supplies a uniform actual row density on K. No tail has been discarded or conditioned away.

On a compact full-dimensional interior simplex $W_* \subset \Delta_{q-1}$ the same Gaussian mixture densities have a uniform positive lower bound. After P this gives $\mu_{a,u} \geq \epsilon_*\zeta$ for the normalized volume law on PW_*, uniformly in u, a. Thus acquired overlap holds with $b = 1$, and it holds for randomized actions as well. Ordinary hyperplane grids supply the global covers and zero boundary mass.

Finally $\bar{L}_a = P$, independently of the sensor. Let $v_* = P(q^{-1}\mathbf{1}) \in K$. Since $0 < (v_*)_i \leq 1$ and $P_{ij} \geq a_0$, for every positive vector z,

$$Pz \geq a_0\Big(\sum_i z_i\Big)v_*.$$

This proves (6) with $\eta_* = a_0$ and $z_* = v_*$. We deliberately choose a minorizing state in $P\Delta$; a uniform vector need not lie there for an arbitrary positive P. All hypotheses are verified without assuming contraction of the optimal sensor selector or a one-step Wasserstein contraction of the filter. □

Example 7.2 (A fixed binary experiment)**.** Take

$$P = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}, \qquad Y \mid I = i \sim N((-1)^i, \sigma_a^2),$$

with any fixed finite nonempty set of positive variances. The Bayes output coordinate is an affine image of a logistic Gaussian, and $K = [1/4, 3/4]$, $D = [1/8, 7/8]$ in the scalar probability chart. Thus the single stationary observer has excess order M^{-2} and the constructed entrance bound has order M^2/n. Here the raw minorization can be improved to $\eta_* = 1/2$ with the uniform state. Scalar Euclidean distance and the categorical task norm differ by a fixed factor $\sqrt{2}$. The experiment is nontrivial even with one sensor; adding sensors enlarges the full policy class. We do not infer that the average-cost optimal sensor switches from a last-step switching example. The actual finite-horizon switching examples and their assumptions remain in the complete controlled companion.

Theorem 7.3 (Noncommuting controlled quantum instruments)**.** *Let physical dimension be $D_q \geq 2$, $0 < \beta < 1$, $0 < \lambda < 1$, and let U_a range over a finite nonempty family of unitary matrices. Let H have normalized Lebesgue measure on the Hilbert–Schmidt unit ball in the traceless Hermitian matrices. The effect, instrument and post-report channel are*

$$E_H = I + \beta H, \quad K_H = E_H^{1/2}, \qquad L_a(H)\sigma = \lambda U_a K_H \sigma K_H U_a^* + (1-\lambda)\operatorname{tr}(E_H\sigma) I/D_q.$$

The action is selected before H. Use a fixed informationally complete finite POVM as the protected audit with categorical squared loss. Then Theorem 3.2 holds with $d = D_q^2 - 1$, in the induced affine probability chart, against all full-history adaptive choices of a. The observer is classical and has no free quantum memory.

Proof. Since $\int H\, d\lambda(H) = 0$, the effects integrate to I. They are positive definite and bounded above and below. The displayed maps are completely positive and integrate to a trace-preserving map. The report density is $\ell(H, \sigma) = 1 + \beta \operatorname{tr}(H\sigma)$, genuinely dependent on the incoming state and bounded between $1 - \beta$ and $1 + \beta$. A fixed informationally complete POVM separates the density matrix, and all future instrument/test expectations are linear in it. This derives the predictive quotient and affine closure; it does not assume a preassigned statistical manifold. The injective affine audit-probability chart carries the actual Brier norm and its affine second moment. All auxiliary Hilbert–Schmidt estimates below transfer through that fixed chart.

Write $\tau = I/D_q$. The physical state base is the compact density body. Take

$$K = \{\sigma \geq (1-\lambda)\tau : \operatorname{tr}\sigma = 1\}, \quad D = \{\sigma \geq \tfrac{1}{2}(1-\lambda)\tau : \operatorname{tr}\sigma = 1\}.$$

Every normalized update is in K, which has a positive buffer in D. The report continuity and integrable operator domination follow from compactness and linearity.

We verify full-dimensional actual density. For fixed interior ρ, the map from a positive trace-D_q effect E to $\sigma = \sqrt{E}\rho\sqrt{E}/\operatorname{tr}(E\rho)$ is a smooth diffeomorphism onto the positive trace-one body. Its inverse is

$$B_{\rho,\sigma} = \rho^{-1/2}(\rho^{1/2}\sigma\rho^{1/2})^{1/2}\rho^{-1/2}, \qquad E = \frac{D_q B_{\rho,\sigma}^2}{\operatorname{tr} B_{\rho,\sigma}^2}.$$

Indeed the positive solution to $B\rho B = \sigma$ is unique; normalization then proves both inverse identities. On $\rho \in D$ and $\|H\|_{\mathrm{HS}} \leq 1$, the effect and the pre-channel posterior have positive uniform spectral floors. The inverse derivative is bounded on this compact set, so the forward Jacobian is bounded away from zero. The final affine map $\sigma \mapsto \lambda U_a \sigma U_a^* + (1-\lambda)\tau$ is invertible on the trace-one

hyperplane. With the bounded actual report density this proves a common upper density bound. Effects with different H need not commute.

For acquired overlap, at $\rho = \tau$ the normalized update is $\tau + (\lambda\beta/D_q)U_a H U_a^*$. Thus the image of a strictly interior effect ball contains a common neighborhood B_* of τ for all actions. The inverse formula and compactness yield a radius $r_* > 0$ and a common positive density on B_* whenever $\|\rho - \tau\|_{\mathrm{HS}} \le r_*$. For arbitrary incoming $\rho \in D$, reports with $\|H\|_{\mathrm{HS}} \le s$ satisfy

$$\|F_a(H,\rho) - \tau\|_{\mathrm{HS}} \le \lambda\|\rho - \tau\|_{\mathrm{HS}} + C\lambda\beta s.$$

To verify this estimate, use $\|\sqrt{I + \beta H} - I\|_{\mathrm{op}} \le C\beta s$, the lower bound $\mathrm{tr}(E_H\rho) \ge 1 - \beta s$, and expand numerator and denominator. Choose $s > 0$ and an integer b so that $C\lambda\beta s/(1-\lambda) < r_*/2$ and $\lambda^{b-1}\sup_D\|\rho - \tau\|_{\mathrm{HS}} < r_*/2$. Each quiet report has actual conditional probability at least $(1-\beta)s^d$, whatever the past actions and reports. After $b-1$ such reports, the final row has the common lower density on B_*. Multiplying these actual conditional probabilities proves (5) uniformly over every feedback policy. All b calls, including the quiet ones, are charged; no conditioning creates a free preparation.

For the different raw condition, integration gives

$$\bar{L}_a z = \lambda U_a \left(\int K_H z K_H \, d\lambda(H) \right) U_a^* + (1-\lambda)\,\mathrm{tr}(z)\tau \ge (1-\lambda)\,\mathrm{tr}(z)\tau.$$

Thus $\eta_* = 1 - \lambda$, $z_* = \tau \in K$. Global affine grids and zero boundary mass complete the verification. The proof is finite-dimensional; no closure statement for unbounded operators is used. □

8. Visible strata and changing predictive rank

A predictive carrier can have full affine dimension while its actual acquisitions occupy smaller pieces. The weights of those pieces belong to the experiment, not to a normalized geometric surrogate.

Theorem 8.1 (Weighted stationary acquired profile)**.** *Replace the full-dimensional row-density condition by the following finite visible decomposition. Each report has a visible mark $j \in \{1, \dots, J\}$ with a fixed actual probability $w_j > 0$, $\sum_j w_j = 1$, independent of the incoming predictive state and action. Conditional on j, its acquired law is supported on a compact convex $K_j \subset K$ in an affine subspace of dimension $d_j \ge 1$, with relative density at most Λ_j, uniformly over $u \in D, a \in A$. Allow additionally finitely many marked atom states. Assume the same acquired overlap, interior buffer, raw minorization and effective boundary-free relative grids. Put*

$$\Xi(M) = \sum_{j : d_j > 0} w_j M^{-2/d_j}. \tag{22}$$

Assume at least one positive-dimensional component. For all M above a fixed threshold depending on the finite decomposition, the stationary optimal excess and checkpoint excess are comparable to $\Xi(M)$. There is one homogeneous observer with finite-acquisition upper

$$R_n - J_*(n) \le C\left(\Xi(M) + \frac{1}{n\Xi(M)}\right),$$

and every M-label observer has lower $c\Xi(M) - C/n$. The upper matches the lower for $n \ge C_*\Xi(M)^{-2}$. Actual weights are retained; constants may depend on the fixed stratum certificates but not on n, M.

Proof. The overlap proof of the Bellman bias does not use full-dimensional density. Strong concavity and a common Lipschitz bound on K remain valid, as does raw ensemble contraction. For the upper reserve one erasure label, one label for each atom, and divide the remaining labels among the finitely many positive-dimensional marked grids. Each gets a fixed positive fraction of M once M exceeds the threshold. Apply the fixed-point correspondence with these marked grids. Every positive-mass cell centroid lies in its convex K_j. The corresponding indicator integrals have closed

graphs because the conditional row charges no relative cell boundary; labels on different marks are kept distinct. The stationary Jensen gap is bounded by

$$C \sum_{d_j>0} w_j m_j^{-2/d_j} + \alpha H \le C\Xi(M) + \alpha H.$$

Atoms have zero cell gap and cause no division by $d_j = 0$. Choose a dyadic erasure rate comparable to $\min\{1/2, \Xi(M)\}$. Ensemble contraction and the preceding proof give the entrance term.

For the converse, any global set of M reconstruction points has conditional distortion at least $c_j M^{-2/d_j}$ on component j. Orthogonal projection of these points into its affine span can only decrease distances, so the relative small-ball proof applies even to off-stratum codepoints. Multiply by the actual w_j and sum. Strong Bellman concavity and (21) give the dynamic lower. The same argument at the acquired random cut proves the checkpoint lower; marked grids give its upper. Mixture weights are neither discarded in a conditioning step nor replaced by weights of a formal reachable union. □

Corollary 8.2 (A raw quantum rank-changing realization)**.** *After the instrument in Theorem 7.3, independently choose a visible mark j with probability w_j and apply a trace-preserving unital conditional expectation $\mathcal{E}_j$ onto a matrix subalgebra. Include the identity map with positive weight; the completely depolarizing map may be one mark. Theorem 8.1 applies with d_j the affine dimension of the trace-one part of that subalgebra, and with an atom for the completely depolarizing mark.*

Proof. The raw marked instrument is $w_j \mathcal{E}_j L_a(H)$ and is completely positive and normalized after summing marks. Each conditional expectation fixes τ and is an orthogonal projection in Hilbert–Schmidt coordinates. It maps K into the compact convex affine section K_j. Projecting the full-dimensional bounded row density of Theorem 7.3, Fubini's theorem bounds the relative density by that upper bound times the maximal volume of the compact fibers. A fixed injective audit chart changes only finite Jacobian constants. The rank-zero map gives exactly τ. Thus each genuine branch mass remains w_j.

The raw minorization is preserved because $\mathcal{E}_j \tau = \tau$. Restricting to the event that all b marks in the quiet-report overlap proof are the identity gives an actual common mass multiplied by $w_{\mathrm{id}}^b > 0$. This event is used as a lower-mass witness, not as an uncharged conditioned experiment. All reports and all marks execute on every run. Relative grids have zero conditional boundary mass. The predictive acquired rank changes between, for example, the full density body and a dephased diagonal body; the physical matrices here remain strictly positive. This does not claim a physical eigenvalue-rank change or extend the theorem to arbitrary pure-state boundaries. The separate pure-rank result is preserved, with its original assumptions, in the Bayesian companion. □

9. Causal simulation, calibration and the same task

Proposition 9.1 (Typed resource and risk composition)**.** *Fix the protected audit protocol, its response space and a loss in $[0, L]$. Suppose a parameter-independent causal simulator from a source experiment to a target has R internal labels, legal strategy lift and action map, causal report transformation, a declared call/physical-time map, and a compatible audit request and task readout. Suppose the total variation distance of the complete stopped and scored laws is at most ε, uniformly over the target policies under comparison. An M-label target observer then gives a source observer with at most RM labels and risk at most its target risk plus $L\varepsilon + e_{mout}$, where e_{mout} is a separately certified expected output-loss perturbation. Serial compatible simulators multiply their label budgets, compose their call maps and add their complete-law errors. A reverse risk lower requires a reverse simulation with its own resource map.*

Proof. Store the pair of simulator and observer labels, route legal actions through the strategy lift, generate each target report without future information, and apply the target update. Independence

from the unknown physical parameter, if any is included in the experiment family, ensures that this is one program. The compatible request fixes the same terminal task; it is not replaced by a model-specific baseline. Integrating the bounded loss against the complete-law difference gives $L\varepsilon$, with total variation defined as the supremum of event differences. Add the output perturbation. Causal composition and the triangle inequality give the serial claim. A one-way inequality has no converse by itself. □

Proposition 9.2 (Stationary perturbation and finite-prefix separation)**.** *Let P be a fixed observer's positive ensemble map satisfying* (20) *with $\gamma_* = \alpha\eta_*$. Let $\widetilde{P}$ be another positive mass-preserving finite-dimensional ensemble map and suppose*

$$\sup_{Z\geq 0,\ e_{\mathrm{tot}}Z=1} \|(P-\widetilde{P})Z\|_{\oplus B} \leq \tau.$$

Every invariant normalized ensemble $\widetilde{z}$ of $\widetilde{P}$ satisfies

$$\|z^* - \widetilde{z}\|_{\oplus B} \leq \frac{\tau}{\gamma_*}. \tag{23}$$

If the perturbed program retains an independent erasure and the same raw minorization, its invariant ensemble is unique and attracts every initial ensemble. Fixed-readout stationary risk differs by at most $L\tau/\gamma_* + e_{mout}$. By contrast a uniform per-step complete-law discrepancy τ gives only a finite-prefix bound $n\tau$ without an additional stability argument; it does not give infinite-transcript total variation convergence.

Proof. Existence of an invariant normalized ensemble follows from the finite-dimensional compact convex base and continuity. Subtract $z^* = Pz^*$ from $\widetilde{z} = \widetilde{P}\widetilde{z}$ and use the contraction of P on equal-mass differences:

$$\|z^* - \widetilde{z}\| \leq (1-\gamma_*)\|z^* - \widetilde{z}\| + \tau.$$

This proves (23). The perturbed raw erasure minorization proves its own attraction as before. The loss functional is bounded affine, so the risk bound follows. The finite-prefix statement follows by sequential coupling or telescoping kernels. It has a factor equal to the actual maximum number of compared calls; no limiting statement is inferred from it. □

Corollary 9.3 (Attained calibration and deficiency in one protected task)**.** *There is a physical extension of every experiment in Theorem 3.2 whose single summed-square task has an unobservable calibration alternative $\pm\delta$ and an attained early-bit deficiency ε, and whose worst-case calibration risk satisfies, for all sufficiently large M,*

$$cM^{-2/d} - C/n + \delta^2 + 2\varepsilon \leq \mathcal{R}^{\mathrm{ext}}_{\mathrm{hom}}(n,M) - J_*(n) \leq C\left(M^{-2/d} + \frac{M^{2/d}}{n}\right) + \delta^2 + 2\varepsilon. \tag{24}$$

The maximum physical-call count is $N = n+3$. In the stationary limit, and also in the matching entrance region, this is a matching resource–resolution–risk law. Numerical and simulator errors are separately added under Propositions 9.1–9.2, not converted automatically into lower bounds.

Proof. After the protected random request, freeze all core acquisition actions. Prepare an independent fair bit $B \in \{-1,1\}$ and send it through a report that reveals B with probability $1-a$ and gives a marked erasure otherwise. This is one paid raw call. Require commitment before the final joint audit, which reveals both the core audit and B. On the binary parameter family of prescribed bit preparations, the deficiency of the erased report relative to the exact bit, with virtual report completion before the audit, is exactly $a/2$. Randomly fill an erased report for the upper; for the lower the sum of the two binary testing errors on an erased report is at least a. Hence put $\varepsilon = a/2$. The fair-bit Bayes squared risk is exactly $a = 2\varepsilon$, using readouts $-1, 0, 1$. The audit cannot be consulted early by a simulator.

The calibration coordinate is an independent physical quantity fixed at either $+\delta$ or $-\delta$, invisible in all preceding reports. Its minimax squared risk is δ^2: read zero for the upper, and average the two alternatives for the lower. It cannot be learned during the core. The extension's loss is the sum of the original audit loss and these two squared losses, so orthogonality/independence makes the lower additive. Conditional on the original observer label, the independent bit report cannot improve core prediction; arbitrary extra randomization does not alter the core lower.

For the upper use m running core labels and, after the request, three readout copies for each of them, corresponding to the bit posterior $-1, 0, 1$. At most $4m + 2$ labels implement all running, terminal and administrative roles. Take m a fixed fraction of M. This is a sufficient construction, not a minimal-state theorem for the enlarged report alphabet. It preserves the core risk order, and the extra coordinate readouts give the exact additive terms. Preparing the core, at most n core reports, the bit call and the single joint audit gives $N = n + 3$. The bit is prepared after controlled actions stop; preparing it earlier would allow later actions to store it in the physical system and would invalidate this lower. The actual calibration alternative and erased-bit channel, rather than permitted error ceilings, are what justify the two added lower terms. □

10. Finite-bit execution and its separate resources

The stationary correspondence proves existence in a hard-label Borel architecture. The following additional statement supplies an effective implementation, without calling the fixed point an algorithm.

Proposition 10.1 (Certified finite compilation)**.** *In addition to Theorem 3.2, assume effective compact nets in K, computable finite instrument integrals, a certified $\eta_* > 0$, and uniform effective boundary/integration moduli for the chosen global grid. Use a fixed dyadic erasure rate α. For any prescribed accuracy parameters there is a finite search over actual M-label programs, with codepoints in a finite σ-net and dyadic action probabilities of denominator Q, whose returned program satisfies*

$$R_\infty \le \rho + C(M-1)^{-2/d} + \alpha H + \frac{L_0 \tau_{\mathrm{tab}}}{\alpha \eta_*} + \kappa + \delta^2_{mout}. \tag{25}$$

Here τ_{tab} is a certified uniform ensemble-map error of table approximation, κ is the final finite search/evaluation slack, and δ^2_{mout} is a certified weighted squared readout error. A finite-precision implementation with per-step ensemble error τ_{mrun} adds at most $L_0\tau_{mrun}/(\alpha\eta_*)$, provided it preserves the independent erasure and the raw minorization. The search leaves the number of retained labels equal to M; it is not claimed to be efficient.

Proof. There is a fixed C_B such that every normalized $z \in C$ is order-dominated by $C_B(1, q)$, uniformly for $q \in K$: a uniform ball about K lies inside the compact state base B. The same interior ball implies, for small $\|q - q'\|$, the two-sided order estimate

$$-C_B\|q - q'\|(1, q) \le_C (0, q' - q) \le_C C_B\|q - q'\|(1, q).$$

Apply the positive instrument, then normalize its mass. Positivity and boundedness of the state base give a common Lipschitz bound for $F_a(y, q)$ in $q \in K$, off the common null set, independent of y. The upper domination also bounds a physical report law at any normalized $z \in C$ by C_B times its law at the programmed codepoint q. Thus codepoint perturbation of size σ changes a cell decision only in a strip of width $C\sigma$ about its grid boundaries, under a report measure dominated by an actual acquired row. The bounded row density gives a uniform strip-mass modulus. For a fixed m-cell grid one may use

$$\tau_{\mathrm{tab}} \le C_m\sigma + C|A|/Q,$$

with a computable finite C_m under the stated grid certificates. More generally the given effective modulus suffices. This bound holds for all positive input ensembles, not only calibrated ones or

positive-mass stationary labels. Rounding action probability vectors to denominator Q changes their ℓ^1 distances by at most $C|A|/Q$.

Enumerate codepoints for all labels from the σ-net and all dyadic action rows; update each by the same report grid and erase with the fixed probability α. Every enumerated object is a genuine positive instrument and has the minorization $\alpha\eta_*$. There are at most

$$|\operatorname{Net}(K,\sigma)|^M (Q+1)^{M|A|}$$

such tables before finite readout rounding. This family includes a table approximating the stationary fixed-point program, whether or not the approximant chooses exact Bellman minimizers. The map error bound and Proposition 9.2 compare their invariant risks with the original fixed readouts.

For each candidate, compute its invariant ensemble by certified integration and positive iteration. After k iterations its exact truncation error is at most $2(1-\alpha\eta_*)^k$, and the integration error has a computable geometric amplification bound. To optimize its readouts, use conditional centroids of this actual invariant ensemble. Very small-mass labels need not be divided by an interval containing zero: if their total certified upper mass is at most $\delta_{\mathrm{out}}^2/L_0$, use any legal response there. For the remaining labels, positive lower mass bounds permit interval division and coordinate refinement to the remaining weighted squared-error budget. Refinement eventually supplies such a partition of labels, including those with zero true mass. The best centered readout does no worse than the unrounded fixed-point readout used for the comparison. Finite interval evaluation and refinement of all candidates yield a program within κ of their minimum. This proves (25).

A running implementation still executes the unmodified physical report kernel. Approximate arithmetic, finite report representation and fallbacks can be coupled to the exact table outside a certified per-step error event. The resulting ensemble-map error is τ_{mrun}; a Gaussian tail fallback does not redefine the physical experiment by truncation. Preserve the exact dyadic erasure coin and use Proposition 9.2. No assertion is made that a program runs forever without any numerical exception. □

To retain the order $M^{-2/d}$, it suffices to choose $\alpha \asymp M^{-2/d}$, $\tau_{\mathrm{tab}} + \tau_{\mathrm{run}} \lesssim \alpha\eta_* M^{-2/d}$, and $\kappa, \delta_{mout}^2 \lesssim M^{-2/d}$. Small mixing gaps are therefore paid for in precision and offline integration, not hidden in label counting. For fixed finite grid rules a table description has the explicit upper form

$$O(Md\log(1/\sigma) + M|A|\log Q + Md\,p_{mout}) + \text{shared primitive and grid description},$$

where p_{mout} is the actually certified readout precision. It is not asserted to be a universal function of δ_{mout} independently of rare-label masses. The retained label needs $\lceil \log_2 M \rceil$ bits. A direct interpreter additionally uses workspace for a d-coordinate primitive evaluation, its precision, a cell search and fresh dyadic coins; its bound is $W_{\mathrm{primitive}}(p) + O(dp + \log M + |A|\log Q)$. Each call takes the certified primitive-evaluation time plus the cell-search and coin costs. Offline enumeration has the displayed finite cardinality and its own iteration/integration time. None of these resources is identified with the persistent label budget, and no computationally optimal lower is claimed.

The compact quantum kernels have effective smooth-integration and strip certificates when their parameters and a fixed interior spectral floor are given effectively. The Gaussian class has the same certificates from its log-odds formula and explicit normal tails on compact parameter certificates. This is an additional effectiveness assertion, not a statement that arbitrary real-valued kernels admit finite descriptions.

11. Comparison, inherited mechanisms and scope

The near-optimality of finite-state controllers for average-cost POMDPs is already an established subject. Yu and Bertsekas [12] prove initial-distribution-independent ϵ-optimal finite controllers for finite state, observation and control spaces when the optimal lower average cost is constant. That existence principle is not a novelty claim of the present paper. Demirci, Kara and Yüksel [2]

derive average-cost optimality and finite-memory/quantized approximations from nonlinear-filter contraction conditions. Our common acquired-overlap condition gives a different sufficient route to a bounded concave bias; it is not a necessary condition for average-cost control.

Saldi, Yüksel and Linder [10] establish finite approximations to Borel-state MDPs, including quantitative and order-optimal results. Our claim is not that finite approximations or rate-optimal control discretization originate here. The missing object addressed here is a homogeneous finite observer of raw reports, with no supplied exact belief at runtime. Its globally calibrated unnormalized ensemble solves all incoming rows and minimizing actions at once. That construction, the acquired-curvature rate, and the all-Borel hard-label converse are the particular combination proved above. Sequential vector quantization [1], value-directed belief compression [7], and Bayes-risk quantization [11] are substantive antecedents, not merely contextual citations.

Recent finite-memory work also requires careful comparison of task quantifiers. Kim [5] explicitly studies a fixed-policy replacement of belief-level costs along the same closed-loop execution. Here $J_*(n)$ and ρ optimize the complete legal acquisition policy, while the constructed controller chooses actions from its own finite labels. Conversely, our finite affine closure, positive actual-row mass and raw minorization are stronger structural restrictions than a general policy-conditional approximation statement. Neither result subsumes the other by a change of terminology.

The predictive realization, likelihood-weighted barycenter identity, acquired-mass curvature estimate and finite-horizon overlap method are inherited and explicitly reproved where used [8, 9]. Kakutani's fixed-point theorem, positive-cone contraction, Bellman span contraction and conditional projection are standard mathematical tools. The new result combines them to close the stationary calibration and all-initial-preparation problem, charges a genuine erasure label, derives an implementable precision region, and verifies two different raw instruments plus a weighted changing-rank extension. It does not infer a universal stationary resource law from finite physical dimension alone.

Several limits are material. The theorem does not characterize all causally compressible experiments. Necessity of acquired overlap, matching optimal transient costs below the entrance region, unknown-kernel learning, arbitrary nondominated or infinite-dimensional realizations, and a complete joint optimum for program, workspace, acquisition time and simulator state are not proved. The protected audit uses an external stopping request; an arbitrary unbounded stopping time or internally generated exact deadline is not covered by that argument. Our stationary perturbation controls bounded task scores, not an infinite-path likelihood ratio or a large-deviation principle. The preserved finite-deadline and singular-boundary results remain valid only under their own stated interfaces and hypotheses.

Appendix A. Three compatibility checks

Example A.1 (A tie must remain a correspondence)**.** If two actions have equal $K_a h(q)$ at a state q, perturbations of q can select opposite strict minimizers. A rule choosing the first minimizer is measurable but need not be continuous. The invariant-ensemble map with such a rule need not have a closed graph. Lemma 5.1 instead allows the entire minimizing simplex; linearity in action probabilities preserves convex values. This is a genuine fixed-point hypothesis check, not a claim that every selected-action map has no fixed point.

Example A.2 (Calibration is not attraction)**.** A Markov map with two absorbing components has many invariant ensembles. Preparing one of them makes its conditional labels calibrated, but another initial component need never approach it. Therefore a calibrated invariant law alone cannot establish an all-initial-law resource theorem. The separate minorization (20) supplies the missing uniqueness and convergence. This elementary example does not satisfy that minorization and is used only to separate the logical statements.

Example A.3 (An external clock is not an internal state)**.** A homogeneous finite observer with equivalent report supports cannot generally reuse the same phase label and also guarantee halting at two different prescribed exact future dates. A possible path from its later occurrence to halting would also be possible from an earlier occurrence. The protected task here instead hides a sampled external request and asks the observer only to read its current label when that request arrives. Its counter belongs to the auditor and is explicitly charged. The stationary theorem cannot be substituted into a total-state exact-deadline formula by treating this counter as free.

These checks complement, rather than remove, the retained local centroid incompatibility, acquired-law versus reachable-set and delayed-information witnesses. None is used as a reason to abandon the general theorem; each identifies the additional hypothesis or accounting step used in its proof.

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